

Two-Stage Fusion for Routing and Classification in Parameter-Efficient Continual Learning

Nguyễn Thiên Trường and Tống Quang Thái

University of Engineering and Technology,
Vietnam National University, Hanoi, Vietnam
24022480@vnu.edu.vn, 24022450@vnu.edu.vn

Abstract

Parameter-efficient continual learning can preserve a pre-trained backbone by assigning compact adapters to incoming tasks. In the task-agnostic setting, however, inference couples two distinct decisions: selecting an adapter and recognising a class among all classes seen so far. These decisions are not equivalent, so a routing correction may leave classification unchanged or disrupt a prediction that was already correct. We address this mismatch with a two-stage fusion framework that augments the adapter-based pipeline with a complementary random-projection (RP) ridge head. The RP head operates in a fixed feature space and is updated from accumulated second-order statistics, without retaining past examples or introducing additional gradient updates to existing adapters. Its scores are reused first to refine the task proposal before re-matching and then to support confidence-aware fusion with the final class logits. Evaluation across multiple datasets, seeds and pre-training settings examines routing, classification and retention separately, showing that the two evidence sources are complementary while their effects remain configuration-dependent.

Keywords: Continual learning · Parameter-efficient tuning · Parameter routing · Random projection · Signal fusion.

1 Introduction

Continual learning asks a model to absorb a stream of tasks without losing what earlier ones taught it. Parameter-efficient tuning adapts a frozen pre-trained backbone using a small number of parameters. One important family maintains task-specific prompts or adapters and selects them at inference [5, 3]. Such methods can avoid storing old examples, but preserving parameters does not guarantee preserved accuracy.

In this family, task identity is unknown at inference and parameter selection is itself a prediction. A wrong adapter can produce unsuitable features. Larger pools, attention-weighted mixtures and explicit re-matching offer different responses to this difficulty [4, 7, 1]. Yet selecting a more suitable adapter and predicting the correct class are different objectives: a shared classifier can recover

from a routing error, while a routing change can also disturb an already correct prediction.

We add a complementary predictor: first-task adapter features, a frozen random projection and a ridge classifier fitted from accumulated statistics, following RanPAC [2]. Fitting uses current-task data without changing LoRA/TII training or retaining examples. Its scores revise the task proposal before re-matching and blend with final class logits, with greater weight when the primary classifier’s top-two margin narrows.

Contributions. We contribute (i) a shared RP predictor for task-proposal and margin-gated class fusion, (ii) matched-feature controls and four-seed conditional ablations, and (iii) a four-dataset, five-pre-training study with weight sensitivity. The novelty is the two-point integration and its evaluation, not ridge learning. We distinguish classification gains from retention and oracle complementarity from achieved corrections.

2 Related Work

Parameter pools and the selection they require. L2P [4] and Dual-Prompt [5] use queries from a frozen backbone to select lightweight parameters from a pool. CODA-Prompt [7] combines prompts using attention weights. HiDe-Prompt [3] separates within-task prediction, task-identity inference and task-adaptive prediction, identifying identity inference as a bottleneck. HRM-PET [1] uses LoRA adapters [6] and re-matching to revisit initial selections. MoTE [14] combines task-specific experts; DLEPEM [15] combines general and adapted prototypes for matching.

Closed-form solutions in continual learning. Closed-form classifiers preserve past contributions through additive statistics of fixed features. APER [8] adapts the backbone on the first task, then uses frozen features and class prototypes. ACIL [20] uses recursive least squares, replacing a sample buffer with correlation statistics and recovering the corresponding joint solution. DS-AL [21] adds a second analytic stream. RanPAC [2] applies a fixed random projection to frozen features, accumulates second-order statistics and solves a ridge system.

We inherit RanPAC’s projection, Gram accumulation and ridge solve, but retain the adapter-routing pipeline and consult the head at two decision points instead of using it as the sole predictor.

Correcting the classifier after the fact. Class-incremental learning often biases predictions towards recent classes [9]. iCaRL [10] uses nearest exemplar means; BiC [11] learns an affine logit correction on held-out data; Weight Aligning [12] matches new- and old-class weight norms; and LUCIR [13] normalises features and weights. Our fusion leaves the adapters, auxiliary classifier and shared head fixed. It retains no exemplars but requires current-task data to fit the RP head.

Remaining challenge. Prompt/classifier inconsistency between training and testing can degrade prediction [18]; re-matching addresses parameter selection but does not supply an independent class predictor. We adopt the re-matching interface of HRM-PET [1], retaining its adapter and classifier training, and test complementary RP evidence before and after re-matching. This differs from replacing adapter selection with an analytic classifier.

3 Method

3.1 Problem Statement

Class-incremental learning (CIL) aims to learn T tasks while mitigating forgetting. Task t provides labelled images $\mathcal{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ from class set $\mathcal{C}_t$, with $\mathcal{C}_t \cap \mathcal{C}_{t'} = \emptyset$ for $t \neq t'$. Only current-task data are available for training; past images are never retained. At inference task identity is unknown and prediction spans all seen classes.

Our task-specific PET setting [3, 1] freezes backbone parameters θ_{ptm} and maintains LoRA adapters $\mathcal{P} = \{p_1, \dots, p_T\}$, one per task, with shared classification head g_{θ_g} . The primary prediction requires adapter selection; our fixed RP branch is independent of the inferred task.

We distinguish four class-score vectors: bare TII scores u , initial adapted shared-head logits r , final re-matched logits ℓ , and RP scores s . The class-to-task map is $\mathcal{T}(\cdot)$. The auxiliary head g_ω maps bare features to u , giving $t_0 = \mathcal{T}(\arg \max_c u_c)$. The initial adapted pass gives $r = g(\phi(x; p_{t_0}))$. DRM evaluates its proposed adapter only if $\hat{t}_{\text{DRM}} \neq t_0$. CRM uses the generalised-entropy confidence of r , tries the task of its second-ranked class, and retains logits according to log-sum-exp; it is not restricted to agreement between task proposals and does not run at task 1. All adapted passes share g ; ℓ denotes the retained class logits. We write $z(\cdot)$ for per-sample standardisation with the population standard deviation floored at 10^{-6} .

3.2 The Second Source of Evidence

We fit an RP head after each task and reuse its scores in two inference-time fusion operations. The existing PET modules receive no additional gradient updates. Figure 1 distinguishes the initial adapted logits from the bare TII scores and shows both reuse paths.

The RP head follows RanPAC [2]. Features $f \in \mathbb{R}^d$, $d = 768$, are column vectors from ViT-B/16 with the first task’s LoRA adapter (p_1 here, index 0 in code), fixed during accumulation and inference and unnormalised; changing adapters between tasks would mix incompatible spaces.

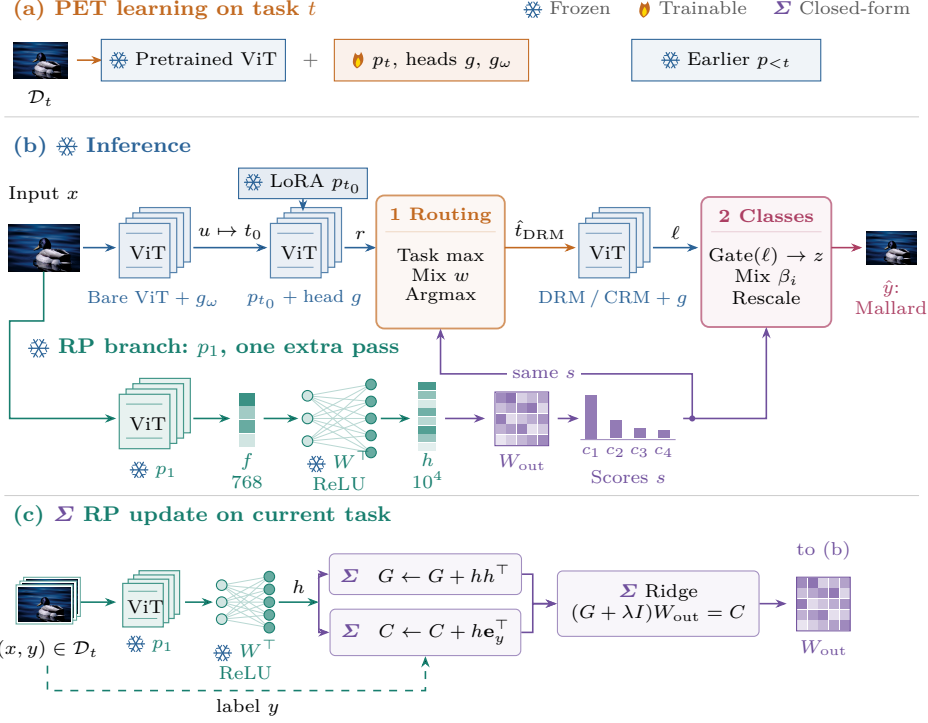


Fig. 1. Parameter lifecycle and two-stage fusion. (a) Training the current PET modules. (b) Fixed RP scores s revise the DRM task proposal and support margin-gated class fusion. (c) Current-task data update G, C, W_{out} using closed-form ridge, without replay or gradient updates.

Projection and second-order statistics. We expand f to $M = 10^4$ dimensions and retain two accumulated matrices:

$$h = \text{ReLU}(W^\top f) \in \mathbb{R}^M, \quad W \in \mathbb{R}^{d \times M}, \quad W_{jk} \sim \mathcal{N}(0, 1), \quad (1)$$

$$G = \sum_n h_n h_n^\top \in \mathbb{R}^{M \times M}, \quad C = \sum_n h_n \mathbf{e}_{y_n}^\top \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad (2)$$

where $\mathbf{e}_y$ is the basis vector of class y . Each current-task training sample contributes once; G and C are accumulated in float64. W is generated once with seed 1993 and then frozen.

The nonlinear expansion permits richer decision functions than a linear head on f . Without ReLU, $W^\top f$ remains linear in f and adds no separating power. This does not guarantee that the expanded data become linearly separable.

Classification by ridge regression. The output weights are the closed-form solution of a regularised least-squares problem:

$$W_{\text{out}} = (G + \lambda I)^{-1} C \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad s = W_{\text{out}}^\top h \in \mathbb{R}^{|\mathcal{Y}|}. \quad (3)$$

The implementation solves the regularised system rather than inverting it.

Two inherited properties. For a fixed feature map, the sums in (2) preserve earlier contributions and yield the joint ridge solution over all observed features, independent of sample order in exact arithmetic. Storage is independent of sample count and retains no per-sample features. This does not imply bitwise floating-point invariance or zero forgetting: new classes can outscore old ones. Changing the first task also changes its adapter and thus the feature map.

3.3 Two-Stage Fusion

Both stages blend sources with different score scales. Direct mixing can make the nominal weights poorly reflect their influence. Each source is standardised per sample:

$$z(v) = \frac{v - \text{mean}(v)}{\sigma_\epsilon(v)}, \quad \sigma_\epsilon(v) = \max\{\text{std}(v), 10^{-6}\}, \quad (4)$$

where std is the population standard deviation. Moments are over seen tasks in Stage 1 and seen classes in Stage 2, excluding unseen entries.

First stage: fusion at the routing level. Stage 1 runs after the initial adapted pass and before DRM. Its input is r , the shared-head logits under the adapter selected by bare TII, not auxiliary logits u . Both r and RP scores s are mapped to seen tasks by

$$r_t^{\text{task}} = \max_{c \in \mathcal{C}_t} r_c, \quad s_t^{\text{task}} = \max_{c \in \mathcal{C}_t} s_c.$$

Here $t = 1, \dots, \hat{T}$, with $\hat{T}$ the number of tasks seen so far; classes of unseen tasks never enter either maximum. The two vectors are blended after standardisation:

$$\begin{aligned} \text{fused} &= w z(r^{\text{task}}) + (1 - w) z(s^{\text{task}}), \quad w = 0.7, \\ \hat{t}_{\text{DRM}} &= \arg \max_t \text{fused}_t. \end{aligned} \quad (5)$$

For one seen task the standard deviation is zero, so the floor keeps the operation finite. The fused task replaces the baseline DRM proposal $\mathcal{T}(\arg \max r)$; the initial TII match, CRM candidate rule, confidence test and shared head are unchanged. At $w = 1$ the ordering is that of r^{task} , recovering that proposal; the full baseline also needs $\beta = 0$.

Second stage: gated fusion at the classification level. Because ℓ comes from the shared head, it can predict the correct class in spite of a wrong routing decision, so we use the projection head itself as a secondary classifier, this time taking its class scores s . A fixed blend would alter confident primary predictions;

we instead reduce the RP share as the top-two probability margin grows, while recognising that confident predictions can still be wrong:

$$\beta_i = \beta \left(1 - \frac{p_{(1)} - p_{(2)}}{p_{(1)} + p_{(2)}} \right), \quad \beta = 0.5, \quad (6)$$

where $p_{(1)} \geq p_{(2)}$ are the two largest probabilities of $\text{softmax}(\ell)$ over the classes seen so far. Dividing by the sum makes the quantity depend only on the ratio $p_{(2)}/p_{(1)}$, so the gate measures how threatened the top class is rather than how much mass the pair holds. The RP share falls to zero as the top class dominates and reaches β when the two are equal: a continuous weight, not a binary test.

The blend reuses (4) and then maps the result back onto the scale of ℓ :

$$m_i = (1 - \beta_i) z(\ell_i) + \beta_i z(s_i), \quad (7)$$

$$\hat{\ell}_i = m_i \cdot \sigma_\epsilon(\ell_i) + \text{mean}(\ell_i) \in \mathbb{R}^{|\mathcal{Y}|}. \quad (8)$$

The positive affine map-back preserves rankings and restores a reference scale for cross-entropy, not a calibration guarantee: for standardised nondegenerate vectors with correlation ρ and the floor inactive, the mixture variance is $(1 - \beta_i)^2 + \beta_i^2 + 2\beta_i(1 - \beta_i)\rho$, which need not equal one. At $\beta = 0$ this stage leaves ℓ unchanged.

Cost. The head adds one adapter forward per sample, shared by both fusion stages. This is not measured total overhead: changing the DRM proposal can also change the number of adapter evaluations.

Per task the head accumulates G and C over the current task’s training split and solves one ridge system. Its stored arrays scale as $O(M^2 + M|\mathcal{Y}| + dM)$: G has M^2 entries, C and W_{out} each have $M|\mathcal{Y}|$, and the projection has dM . Storage does not grow with the number of observed samples for fixed dimensions and class capacity. The Gram matrix dominates in our setting. At $M = 10^4$ it holds 10^8 float64 entries, or 800 MB. Together with its regularised copy and Cholesky factor, these three matrices require approximately 2.4 GB; this is an analytical estimate, not a measured total-memory peak. A historical ten-task evaluation on ImageNet-R took 283 s on one RTX 4090 at batch 24; neither its phase-level costs nor its overhead over the baseline were measured.

4 Experiments

4.1 Setup

The backbone is ViT-B/16 [22], pre-trained on ImageNet-21K. Rank-8 LoRA modifies the key/value projections in the first five blocks. The four-seed study splits ImageNet-R [23], CIFAR-100 [24] and CUB-200-2011 [25] into 10 tasks of 20, 10 and 20 classes, respectively. The grid replaces CUB with ImageNet-A [26] (10×20 classes) and the five-dataset protocol of [1] (one domain per task). Its

pre-training settings are Sup-21K, MoCo-v3 [27], iBOT-1K/iBOT-21K [28] and DINO [29].

Each seed (42–45) has separately trained LoRA/TII parameters; pre-trained backbone weights remain frozen. Both methods use identical saved checkpoints and evaluation paths. RP fits current-task training images once using the evaluation transform: resize to 256, centre crop to 224, and convert to a tensor; CIFAR additionally uses its configured channel normalisation. No test labels enter the fit. We use $M = 10^4$, ReLU, $\lambda = 10^4$, no additional RP feature normalisation or temperature calibration, and no fusion ramp. Unless noted otherwise, we report $w = 0.7$ and $\beta = 0.5$ throughout, not a dataset-wise best configuration. The joint sensitivity study in Sect. 4.6 includes the better observed $w = 0.6$ setting; we do not retroactively select the main configuration on that test grid. Fusion starts at task 1; CRM retains **En=gen**, threshold -10 , and the second-ranked class candidate. Training/evaluation scripts and dataset preprocessing are versioned in the code repository [30].

Task-routing accuracy in the released evaluator measures the DRM task proposal. In the default class-based routing path, CRM can change the output logits without updating the task identity passed to this metric; it is therefore not a measure of final adapter selection after CRM. Top-1 and Top-5 classification accuracy, as well as Loss, use the final classification logits. Final metrics average the per-task values with equal task weights, not by pooling all test images. Writing $a_{i,t}$ for classification accuracy on task i at stage t , the retention measures are

$$\text{Forgetting} = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(\max_{i \leq t \leq T} a_{i,t} - a_{i,T} \right), \quad (9)$$

$$\text{Backward} = \frac{1}{T-1} \sum_{i=1}^{T-1} (a_{i,T} - a_{i,i}). \quad (10)$$

Both averages exclude the last task, and the maximum includes only stages at which task i has been learned. Lower is better for Loss and Forgetting; higher is better for accuracy and Backward (including when Backward is negative).

Reproducibility and sanity check. With corrected unseen-class masking, $w = 1, \beta = 0$ recovers the unfused path to logged precision at every stage for all three datasets and seeds 42–45. For example, both paths give ImageNet-R seed-42 Top-1 74.0191. This identity does not imply harmless intermediate fusion settings. Main runs, controls, grids, ablations and diagnostics were re-evaluated; corrected loss/retention values are used throughout. All forty grid evaluations and fixed backbone/first-adapter tensors were checked.

4.2 Main Results

Table 1 combines published context and our controlled comparisons. Its groups have different seed sets and, in the last group, different pretrained weights; they

Table 1. Main results: final task-averaged Top-1 (%). Sup-21K groups use the paper backbone; the last group uses RanPAC’s AugReg checkpoint. Four-seed values are mean $\pm$ sample SD; single-run values have no SD. Dashes mean not evaluated/reported. IMR/IMA denote ImageNet-R/A; CIFAR denotes CIFAR-100. Bold compares only the two matched methods within each middle group, not all protocols.

Method	IMR	CIFAR	IMA	5-Datasets
<i>Published Sup-21K context: three-seed means from [1]</i>				
DualPrompt [5]	68.79	86.60	39.76	76.09
S-Prompt++ [16]	69.68	88.81	39.10	83.38
CODA-Prompt [7]	70.03	86.94	30.96	62.54
LAE [17]	72.25	85.10	40.36	64.16
CPrompt [18]	69.24	83.18	37.60	71.72
InfLoRA [19]	72.24	85.28	42.12	71.97
HiDe-LoRA [3]	71.90	88.43	42.18	93.25
HRM-PET [1]	73.86	89.45	44.28	93.99
<i>Our matched Sup-21K runs: seeds 42–45 (IMR and CIFAR only)</i>				
HRM-PET	73.94 $\pm$ 0.48	89.78 $\pm$ 0.03	–	–
Full (ours)	75.18 $\pm$ 0.24	90.47 $\pm$ 0.05	–	–
<i>Our Sup-21K grid: seed 42 (overlaps the four-seed group)</i>				
HRM-PET	74.02	89.77	45.85	91.37
Full (ours)	75.51	90.53	47.42	93.77
<i>Exploratory AugReg-21K$\rightarrow$1K: order 1993, training seed 1</i>				
RanPAC [2]	77.44	–	–	–
HRM-PET	77.42	–	–	–
Full (ours)	79.14	–	–	–

must not be pooled into a single ranking. Published values are quoted from Table 1 of [1], not rerun here. S-Prompt++ and HiDe-LoRA are the variants evaluated in that benchmark.

For the four-seed comparisons, differences are computed within each seed before averaging. Figure 2 shows paired 95% Student- t intervals, $\bar{\Delta} \pm 3.182 s_{\Delta}/\sqrt{4}$, using sample SD. With four seeds these intervals are imprecise and assume approximately normal seed-wise differences; correlated metrics are not independent confirmations.

Prediction and retention. Task routing rises by 1.10–2.26 points and Top-1 by 0.69–1.50, with positive paired intervals on all three datasets. Top-5 also rises by 0.22–1.31 points, with positive intervals. The Top-1 gain repeats in every seed, but these aggregate gains do not count how many routing corrections repair a class prediction. All six paired retention intervals include zero: neither a consistent benefit nor equivalence is established. For example, CIFAR Forgetting changes by -0.150 [$-0.302, +0.002$] and Backward by $+0.136$ [$-0.015, +0.287$]. No equivalence margin was specified.

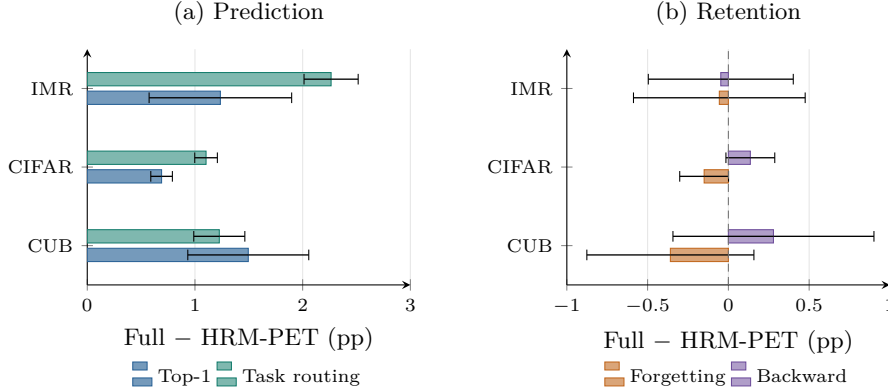


Fig. 2. Paired changes over seeds 42–45; whiskers are 95% t intervals, not the SD of individual runs. Positive is better for prediction and Backward; negative is better for Forgetting. Every retention interval crosses zero. IMR, CIFAR and CUB denote ImageNet-R, CIFAR-100 and CUB-200.

Comparability. Our seed-42 HRM-PET accuracy differs from the published three-seed mean by +0.16 on ImageNet-R, +0.32 on CIFAR-100, +1.57 on ImageNet-A and -2.62 on 5-Datasets. Published-context rows are therefore not controlled comparisons with our runs; the paired differences isolate the added fusion.

The last group uses the same AugReg weights (ImageNet-21K followed by ImageNet-1K fine-tuning), hashed images and class order (NumPy seed 1993). RanPAC uses its official ID7 entry (SSF, 20 epochs on task 1); our TII/LoRA training uses 20/50 epochs per task. Torch seed is 1, but augmentations, run-times and training budgets differ. Full gains 1.69 points over RanPAC and 1.72 over matched HRM-PET. This one-order result establishes neither multi-order reliability nor lower computational cost; pending replications are excluded.

4.3 Generalisation Across Pre-training Settings

Table 2 reports the 4×5 grid using one code path at seed 42. All twenty cells improve Top-1 by +0.59 to +3.58 points, against baseline Top-1 of 31.8–93.8; task-routing gains are +0.97 to +4.57.

Final accuracy on earlier tasks rises in every cell (+1.83 on average), versus +1.58 at learning time. Backward, their difference, worsens in five cells: learning-time accuracy rose more, not final accuracy less. This decomposition uses rounded per-task logs and remains descriptive.

4.4 Ablation

Table 3 isolates the matched-feature RP predictor and the class-fusion variants. All RP variants use the same fixed first-task features and ridge configuration. RP-only predicts directly from s and is not a reproduction of the complete RanPAC protocol. Class-only disables routing fusion ($w = 1$); both class-only variants use $\beta = 0.5$.

Table 2. Top-1 change (points), full minus matched HRM-PET at seed 42. All cells use $w = 0.7$, $\beta = 0.5$ and the margin gate; no confidence intervals are implied.

Pre-training	ImageNet-R	CIFAR-100	ImageNet-A	5-Datasets
Sup-21K	+1.49	+0.76	+1.57	+2.40
MoCo-v3	+1.89	+2.12	+2.41	+1.41
iBOT-1K	+2.52	+2.06	+3.58	+0.85
iBOT-21K	+2.55	+1.80	+2.63	+0.59
DINO	+2.68	+2.03	+0.97	+0.75

Table 3. Final Top-1 (%), mean $\pm$ sample SD over seeds 42–45. Full uses $w = 0.7$, $\beta = 0.5$ and the margin gate. Bold marks the highest column mean, not significance; full is not uniformly best.

Variant	ImageNet-R	CIFAR-100	CUB-200
HRM-PET	73.94 $\pm$ 0.48	89.78 $\pm$ 0.03	86.38 $\pm$ 0.25
Same-feature RP-only	69.89 $\pm$ 0.17	88.80 $\pm$ 0.20	87.26 $\pm$ 0.27
Class-only, ungated	74.49 $\pm$ 0.14	90.15 $\pm$ 0.16	88.01 $\pm$ 0.21
Class-only, gated	75.10 $\pm$ 0.21	90.50 $\pm$ 0.08	87.72 $\pm$ 0.26
Full (ours)	75.18 $\pm$ 0.24	90.47 $\pm$ 0.05	87.88 $\pm$ 0.27

Full exceeds RP-only on all three datasets, but class fusion accounts for much of the gain over HRM-PET. Ungated class-only has the highest CUB mean and gated class-only the highest CIFAR mean; we retain one configuration rather than select a different winner on each dataset.

Routing with ungated class fusion. The five-backbone ImageNet-R ablation uses seed 42; baseline Top-1 ranges from 68.47 to 74.02. We report differences between specified variants. Routing-only and ungated class-only variants improve Top-1 on each backbone by +0.29–+0.73 and +0.51–+3.07, respectively. Adding routing to ungated class fusion changes Top-1 by -0.07 to $+0.02$, not establishing equivalence.

Full-minus-gated-class-only Top-1 is +0.13, +0.29, +0.78, +0.43 and +0.77 on Sup-21K, MoCo-v3, iBOT-1K, iBOT-21K and DINO at seed 42. These single-seed gains establish neither reliability nor retention benefit: MoCo-v3 Forgetting rises from 3.44 to 3.69. The gate is not uniformly beneficial either: full minus ungated full is -0.26 Top-1 points on MoCo-v3 and -0.54 on iBOT-1K.

Conditional contributions over four seeds. Full minus gated class-only measures conditional routing; gated minus ungated class-only measures the gate without routing fusion. These are not independent stage effects.

The conditional routing effect is $+0.083$ [$+0.026, +0.140$] on ImageNet-R, -0.030 [$-0.085, +0.025$] on CIFAR-100 and $+0.156$ [$+0.055, +0.257$] on CUB-200. The gate is $+0.603$ and $+0.355$ on the first two, both excluding zero, and -0.284 [$-0.570, +0.002$] on CUB-200. Each pair sums to the full-against-ungated-class-only change, $+0.69$, $+0.33$ and -0.13 , to within 0.005. Routing is positive on two datasets and unresolved on CIFAR. The gate’s CUB estimate is negative but its interval includes zero; this does not establish harm. Full still improves over HRM-PET in Table 1. Only the first fusion stage changes task routing.

4.5 Complementarity Diagnostics

These seed-42 diagnostics use equal task weights (Sect. 4.1). On ImageNet-R, bare TII routing accuracy is 63.80%, the baseline DRM proposal 77.79% and the RP router 77.12%. The TII-RP oracle union reaches 82.99%. Dividing the RP-correct/TII-wrong rate by TII’s error rate gives 53.0%, versus 62.0% on CIFAR-100 and 46.9% on CUB-200. These are task-balanced recovery ratios, not pooled image counts or achieved fusion accuracy. They compare bare TII with RP, not Stage 1’s adapted logits with RP.

Complementarity also exists at the class level. In the routing-only diagnostic ($w = 0.7$, $\beta = 0$), ImageNet-R accuracy is 74.31% for the routed head and 70.08% for RP, with an oracle union of 78.89%. RP is correct when the routed head is wrong on 4.58 task-balanced percentage points, versus 2.43 on CIFAR-100 and 3.87 on CUB-200. This potential motivates reusing s for classification, but does not count actual repairs or harms of full fusion.

4.6 Sensitivity to the Mixing Weights

The two weights were fixed sequentially, w before the second stage existed and β afterwards, so the second stage may have shifted the optimum of the first. Figure 3 plots the joint grid on ImageNet-R at seed 42 with the margin gate active. This is an exploratory test-set sensitivity analysis, not an independent validation set for choosing hyperparameters.

The grid favours $\beta = 0.5$ at every w ; its maximum is $w = 0.6$, not the reported 0.7. At $\beta = 0.5$, Top-1 spans 0.11 points, versus about one point for task routing. Across four seeds, 0.6 gains $+0.332$ [$+0.143, +0.522$] task-routing points and $+0.108$ [$-0.064, +0.280$] Top-1 points over 0.7 with the gate and $\beta = 0.5$; the Top-1 sign reverses on seed 43. Thus 0.6 is the observed grid maximum, while its classification advantage over 0.7 is small and uncertain across four seeds. We retain 0.7 as the common reporting configuration, not as an optimum, and disclose the full sensitivity result rather than selecting the main setting after inspecting these test outcomes. This does not remove the development-set bias.

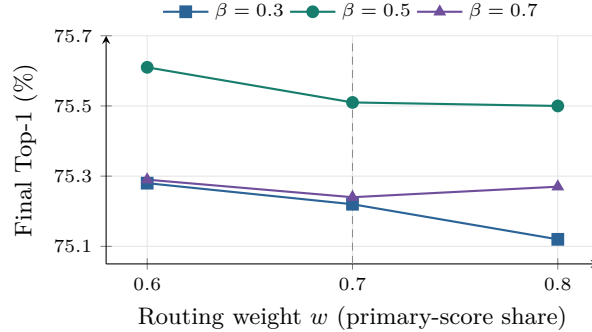


Fig. 3. Weight sensitivity, ImageNet-R seed 42. The displayed vertical range is zoomed to show small differences; points are individual runs, not means or confidence intervals. The dashed line marks the reported $w = 0.7$. The best observed grid point is $(w, \beta) = (0.6, 0.5)$ at 75.61%.

4.7 Limitations

ImageNet-R and CUB informed development; neither is an untouched holdout. More seeds do not remove selection bias. CUB/pre-training overlap [25] was not audited. The grids and five-backbone ablation are single-seed; only the three-dataset contrasts use four seeds. CIFAR’s conditional routing effect is unresolved, and gate benefits vary across backbones. The additional RanPAC comparison matches pretrained weights, images and class order only for one run; training budgets and runtimes differ, and multi-order verification is incomplete. The RP-only control in Table 3 is not a full RanPAC reproduction. Quoted baselines use another implementation; our Sup-21K 5-Datasets result is below the published figure. Retention benefit/equivalence remains unestablished; phase-level cost and actual full-fusion repair/harm counts were not measured. The loader skips two zero-byte NotMNIST training files, also empty in the local source archive; we did not replace them.

5 Conclusion

Reusing one fixed-feature RP predictor for task proposals and class scores improves classification by 0.69–1.50 Top-1 points over the matched PET system across three datasets and four seeds, without changing adapter training or storing past images. Component comparisons show that class fusion provides much of the gain; the added value of routing and gating is conditional, not universal. The method improves prediction but does not establish reduced forgetting. Controlled cost measurements and independent validation are priorities for further evaluation.

Acknowledgment. We thank our supervisor for comments on this manuscript.

References

1. Weicheng Wang, Guoli Jia, Xialei Liu, Liang Lin, Jufeng Yang: Hybrid re-matching for continual learning with parameter-efficient tuning. In: NeurIPS (2025)
2. Mark D. McDonnell, Dong Gong, Amin Parvaneh, Ehsan Abbasnejad, Anton van den Hengel: RanPAC: random projections and pre-trained models for continual learning. In: NeurIPS (2023)
3. Liyuan Wang, Jingyi Xie, Xingxing Zhang, Mingyi Huang, Hang Su, Jun Zhu: Hierarchical decomposition of prompt-based continual learning: rethinking obscured sub-optimality. In: NeurIPS (2023)
4. Zifeng Wang, Zizhao Zhang, Chen-Yu Lee, Han Zhang, Ruoxi Sun, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, Tomas Pfister: Learning to prompt for continual learning. In: CVPR, pages 139–149 (2022)
5. Zifeng Wang, Zizhao Zhang, Sayna Ebrahimi, Ruoxi Sun, Han Zhang, Chen-Yu Lee, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, Tomas Pfister: DualPrompt: complementary prompting for rehearsal-free continual learning. In: ECCV, pages 631–648 (2022)
6. Edward Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, Weizhu Chen: LoRA: low-rank adaptation of large language models. In: ICLR (2022)
7. James Seale Smith, Leonid Karlinsky, Vyshnavi Gutta, Paola Cascante-Bonilla, Donghyun Kim, Assaf Arbelle, Rameswar Panda, Rogerio Feris, Zolt Kira: CODA-Prompt: continual decomposed attention-based prompting for rehearsal-free continual learning. In: CVPR, pages 11909–11919 (2023)
8. Da-Wei Zhou, Zi-Wen Cai, Han-Jia Ye, De-Chuan Zhan, Ziwei Liu: Revisiting class-incremental learning with pre-trained models: generalizability and adaptivity are all you need. *International Journal of Computer Vision*, volume 133, number 3, pages 1012–1032 (2025). <https://link.springer.com/article/10.1007/s11263-024-02218-0>
9. Marc Masana, Xialei Liu, Bartłomiej Twardowski, Mikel Menta, Andrew D. Bagdanov, Joost van de Weijer: Class-incremental learning: survey and performance evaluation on image classification. *Institute of Electrical and Electronics Engineers Transactions on Pattern Analysis and Machine Intelligence*, volume 45, number 5, pages 5513–5533 (2023)
10. Sylvestre-Alvise Rebuffi, Alexander Kolesnikov, Georg Sperl, Christoph H. Lampert: iCaRL: incremental classifier and representation learning. In: CVPR, pages 2001–2010 (2017)
11. Yue Wu, Yinpeng Chen, Lijuan Wang, Yuancheng Ye, Zicheng Liu, Yandong Guo, Yun Fu: Large scale incremental learning. In: CVPR, pages 374–382 (2019)
12. Bowen Zhao, Xi Xiao, Guojun Gan, Bin Zhang, Shu-Tao Xia: Maintaining discrimination and fairness in class incremental learning. In: CVPR, pages 13208–13217 (2020)
13. Saihui Hou, Xinyu Pan, Chen Change Loy, Zilei Wang, Dahua Lin: Learning a unified classifier incrementally via rebalancing. In: CVPR, pages 831–839 (2019)
14. Linjie Li, Zhenyu Wu, Yang Ji: MoTE: mixture of task-specific experts for pre-trained model-based class-incremental learning. *Knowledge-Based Systems*, volume 324, article 113795 (2025). <https://doi.org/10.1016/j.knosys.2025.113795>
15. Hongwei Zhao, Rui Liu, Yansong Liu: Dynamic LoRA-experts and prototype-ensemble matching for class-incremental learning. *Applied Sciences*, volume 16, number 12, article 6153 (2026). <https://doi.org/10.3390/app16126153>

16. Yabin Wang, Zhiwu Huang, Xiaopeng Hong: S-Prompts learning with pre-trained transformers: an Occam’s razor for domain incremental learning. In: NeurIPS (2022)
17. Qiankun Gao, Chen Zhao, Yifan Sun, Teng Xi, Gang Zhang, Bernard Ghanem, Jian Zhang: A unified continual learning framework with general parameter-efficient tuning. In: ICCV, pages 11483–11493 (2023)
18. Zhanxin Gao, Jun Cen, Xiaobin Chang: Consistent prompting for rehearsal-free continual learning. In: CVPR, pages 28463–28473 (2024)
19. Yan-Shuo Liang, Wu-Jun Li: InfLoRA: interference-free low-rank adaptation for continual learning. In: CVPR, pages 23638–23647 (2024)
20. Huiping Zhuang, Zhenyu Weng, Hongxin Wei, Renchunzi Xie, Kar-Ann Toh, Zhiping Lin: ACIL: analytic class-incremental learning with absolute memorization and privacy protection. In: NeurIPS (2022)
21. Huiping Zhuang, Run He, Kai Tong, Ziqian Zeng, Cen Chen, Zhiping Lin: DS-AL: a dual-stream analytic learning for exemplar-free class-incremental learning. In: AAAI, volume 38, number 15, pages 17237–17244 (2024). <https://ojs.aaai.org/index.php/AAAI/article/view/29670>
22. Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiuhua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby: An image is worth 16x16 words: transformers for image recognition at scale. In: ICLR (2021). <https://arxiv.org/abs/2010.11929>
23. Dan Hendrycks, Steven Basart, Norman Mu, Saurav Kadavath, Frank Wang, Evan Dorundo, Rahul Desai, Tyler Zhu, Samyak Parajuli, Mike Guo, Dawn Song, Jacob Steinhardt, Justin Gilmer: The many faces of robustness: a critical analysis of out-of-distribution generalization. In: ICCV, pages 8340–8349 (2021). <https://github.com/hendrycks/imagenet-r>
24. Alex Krizhevsky: Learning multiple layers of features from tiny images. Technical report, University of Toronto (2009). <https://cave.cs.toronto.edu/kriz/learning-features-2009-TR.pdf>
25. Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, Serge Belongie: The Caltech-UCSD Birds-200-2011 dataset. Technical report number CNS-TR-2011-001, California Institute of Technology (2011). <https://authors.library.caltech.edu/records/cvm3y-5hh21>
26. Dan Hendrycks, Kevin Zhao, Steven Basart, Jacob Steinhardt, Dawn Song: Natural adversarial examples. In: CVPR, pages 15262–15271 (2021)
27. Xinlei Chen, Saining Xie, Kaiming He: An empirical study of training self-supervised vision transformers. In: ICCV, pages 9640–9649 (2021)
28. Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, Tao Kong: iBOT: image BERT pre-training with online tokenizer. In: ICLR (2022). <https://arxiv.org/abs/2111.07832>
29. Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, Armand Joulin: Emerging properties in self-supervised vision transformers. In: ICCV, pages 9650–9660 (2021)
30. Nguyễn Thiên Trường, Tống Quang Thái: Two-stage fusion for PET-based continual learning: implementation, paper-system revision 4989ab8 (2026). <https://github.com/Truong140206/Two-Stage-Fusion-for-PET-Based-Continual-Learning->